

Bridging the Gap Between Simulation and Reality for Ultra-Reliable Communications

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Abstract—Constructing accurate spatial reliability maps is essential for Ultra-Reliable and Low-Latency Communication (URLLC). Yet achieving this remains challenging: maps built from simulation suffer from a significant sim-to-real gap, while maps from scarce real-world measurements fail to capture the rare, low-SINR extreme events that govern tail reliability. In this letter, we propose a novel domain-adaptive, extreme value theory-based framework to construct high-fidelity reliability maps. We use a teacher–student scheme, where the teacher learns a generic tail model from simulation using a Generalized Pareto Distribution (GPD) objective, and the student is subsequently adapted to the real environment using a composite loss incorporating real-sample GPD characteristics, teacher-guided distillation, and feature-alignment terms. We validate our model using a real-world indoor testbed and a corresponding ray-tracing simulation. Results show our approach successfully bridges the sim-to-real gap, reducing the 99.9% target SINR prediction error by over 55%, confirming that our method is robust and particularly well-suited for stringent URLLC reliability requirements.

I. INTRODUCTION

ULTRA-RELIABLE and Low-Latency Communication (URLLC) has been envisioned as one of the three cornerstone service classes of 5G to support mission-critical applications such as industrial automation, autonomous driving, and remote healthcare [11]. These applications demand stringent performance targets [8], e.g., end-to-end latency on the order of 1 ms and block-error rates below 10^{-3} , in order to ensure dependable service delivery. However, achieving such targets in today’s 5G networks remains challenging. In indoor scenarios, severe multipath propagation and local scattering lead to pronounced spatial variations and temporal fluctuations of the Signal-to-Interference-plus-Noise Ratio (SINR), which can hinder timely power control and modulation-and-coding-scheme (MCS) adaptation.

To quantify the impact of the wireless environment, radio maps are a vital tool for network planning, providing spatial distributions of key metrics like SINR. Earlier work on statistical radio maps [12] proposed to estimate channel distributions non-parametrically from empirical samples, constructing spatial maps of outage capacity to enable URLLC rate selection without assuming any specific channel model. However, these and other existing methods are inadequate for URLLC. Maps based on high-fidelity ray-tracing simulators [3] are scalable but suffer from an unavoidable *sim-to-real gap*, failing to capture the uncertainties of real deployments.

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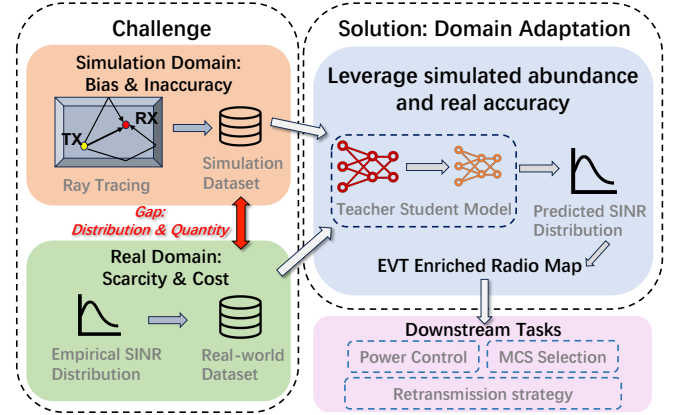


Fig. 1: Challenges and our solution

While some studies incorporate Extreme Value Theory (EVT) to model tail behavior [5], [7], their reliance on simulation limits their real-world accuracy. Conversely, methods trained on real-world measurements [6] can learn the environment but fundamentally fail to model the rare, low-SINR *extreme events* that define URLLC reliability.

This paper addresses the central question: *can we create a reliability map that combines the scalability of simulation with the accuracy of real-world measurements, while also capturing the extreme-value behavior critical for URLLC?* This requires solving two fundamental challenges showed in Figure 1: **(C1) The Scarcity of Extreme Events**, as rare low-SINR tail samples are difficult to capture in practice, and **(C2) The Sim-to-Real Gap**, as simulated data distributions do not match reality.

In this letter, we propose a novel domain-adaptive, EVT-based framework to construct high-fidelity reliability maps. We employ a teacher-student architecture where a teacher model learns the general tail distribution from abundant simulation data, and a student model adapts this knowledge to a small set of real-world measurements using a composite loss function based on Negative Log-Likelihood (NLL). Our main contributions are as follows:

- We design a domain-adaptive learning framework that leverages abundant simulated data while being anchored by a small set of real-world measurements.
- We incorporate extreme value theory directly into the NLL loss function to accurately characterize the low-SINR tail distribution.
- We demonstrate through real-world traces that our method bridges the sim-to-real gap, producing reliable EVT-based radio maps and significantly outperforming

simulation-only baselines.

II. SYSTEM MODEL

We consider a single-cell indoor industrial environment where a Base Station (BS) serves a User Equipment (UE) operating under URLLC requirements. The instantaneous SINR experienced by the UE at location l is denoted by $\gamma(l)$. Due to multipath propagation and interference, $\gamma(l)$ fluctuates randomly over time even for a fixed position.

For URLLC, the *tail behavior* of $\gamma(l)$ is critical. We define the outage probability at location l as

$$O(l) = P\{\gamma(l) < \gamma_{\text{tar}}\} \leq \zeta, \quad (1)$$

where ζ is the predefined reliability target (e.g., 10^{-3}) and γ_{tar} is the required SINR for that target.

Spatially modeling $O(l)$ across the deployment area yields what we term the **reliability map**—a spatial representation of link reliability that quantifies the probability of meeting URLLC requirements at each location. An accurate reliability map enables proactive resource allocation and risk-aware link adaptation, ensuring that ultra-reliable performance is consistently achievable throughout the environment.

A. EVT-Based Tail Modeling

In practice, the full distribution of $\gamma(l)$ is unknown or difficult to estimate from limited samples. Extreme Value Theory (EVT) provides a rigorous framework for characterizing the statistical behavior of the tails of random variables, independent of their original distributions. According to the theorem of exceedances over thresholds [9], for a random variable X drawn from a non-degenerate distribution and for a sufficiently high threshold μ , the conditional distribution of the excess variable $Z = X - \mu \mid X > \mu$ converges to the Generalized Pareto Distribution (GPD):

$$F_Z(z) = 1 - (1 + \xi z/\sigma)^{-1/\xi}, \quad z > 0, \quad 1 + \xi z/\sigma > 0, \quad (2)$$

where $\mu, \sigma > 0$, and ξ denote the threshold, scale, and shape parameters, respectively.

In our context, we define $X = f(\gamma(l))$ with a monotonic transformation $f(\cdot)$ (e.g., $f(x) = -x$) to map low SINR values (the left tail) to high values (the right tail) suitable for GPD modeling. The excess data above the threshold $\mu(l)$ are then given by

$$\psi(l) = f(\gamma(l)) - \mu(l) \mid f(\gamma(l)) > \mu(l), \quad (3)$$

from which the parameters $\sigma(l)$ and $\xi(l)$ are estimated. The threshold $\mu(l)$ is typically determined by a quantile rule [10] $\mu(l) = Q(\rho \times 100, f(\gamma(l)))$, where ρ represents the upper-tail fraction (e.g., $\rho = 0.9$).

Substituting these estimates into the GPD model yields the approximate outage probability:

$$O(l) = (1 - \rho) \left(1 + \frac{\hat{\xi}(l)}{\hat{\sigma}(l)} (f(\gamma_{\text{tar}}) - \hat{\mu}(l)) \right)^{-1/\hat{\xi}(l)}. \quad (4)$$

Equation (4) establishes an analytical relation between the URLLC outage requirement and the statistical tail parameters of $\gamma(l)$.

B. Target SINR Derivation

Given the desired reliability constraint $O(l) = \zeta$, we can invert (4) to compute the *target SINR threshold* that satisfies the URLLC requirement:

$$\gamma_{\text{tar}}(l) = f^{-1} \left(\hat{\mu}(l) + \frac{\hat{\sigma}(l)}{\hat{\xi}(l)} \left[\left(\frac{\zeta}{1 - \rho} \right)^{-\hat{\xi}(l)} - 1 \right] \right). \quad (5)$$

Intuitively, $\gamma_{\text{tar}}(l)$ represents the SINR level that must be exceeded with probability $1 - \zeta$ at location l . For example, if $\zeta = 10^{-2}$, then $\gamma_{\text{tar}}(l)$ corresponds to the 1% lower-tail quantile of the local SINR distribution—the value below which SINR falls only 1% of the time. In practice, this implies that to meet stricter reliability demands, the system must select a more robust modulation and MCS or a lower channel quality indicator (CQI) corresponding to $\gamma_{\text{tar}}(l)$.

This formulation directly links the reliability constraint ζ to a spatially varying SINR threshold, enabling the construction of *tail-aware SINR threshold maps*. These maps serve as the foundation for learning-based spatial generalization and domain adaptation (DA) across simulated and real-world environments.

III. PROPOSED DOMAIN ADAPTATION DL MODEL

A. Motivation and Problem Formulation

To bridge the simulation-to-reality discrepancy, we propose a deep learning-based DA model that transfers the tail-distribution knowledge learned from simulation to the real environment.

Let $D_s = \{(x_i^s, \mu_i^s, \Gamma_i^s)\}_{i=1}^{N_s}$ and $D_r = \{(x_j^r, \mu_j^r, \Gamma_j^r)\}_{j=1}^{N_r}$ denote the datasets from the simulated and real domains, respectively, where x represents spatial and environmental features (e.g., coordinates, materials, and line-of-sight conditions), Γ is the set of raw SINR samples (either simulated or real) at location x , and μ is the pre-calculated threshold for that location. The model's objective is to learn a mapping $f_\theta : x \rightarrow y$ that accurately predicts the GPD tail parameters $y = (\sigma, \xi)$ in the real domain. Typically, $N_s \gg N_r$ due to the scarcity of measured data.

B. Teacher-Student Architecture

Inspired by the DA framework in [4], we design a two-stream *teacher-student* neural network, as illustrated in Figure 2. The teacher network f_{θ_1} is first trained on abundant simulated samples, while the student network f_{θ_2} is trained on scarce real samples with the assistance of the teacher. Both networks share the same multilayer perceptron (MLP) architecture but have independent parameters.

Teacher Pre-Training Objective: The simulation dataset $D_s = \{(x_i^s, \mu_i^s, \Gamma_i^s)\}_{i=1}^{N_s}$ contains raw simulated SINR samples Γ_i^s and pre-calculated thresholds μ_i^s , analogous to the real-world dataset. The teacher is obtained by minimizing the GPD Negative Log-Likelihood (NLL) [15] loss over the simulation dataset:

$$\mathcal{L}_{\text{teacher}}(\theta_1) = \mathbb{E}_{(x, \mu, \Gamma) \sim D_s} \left[\sum_{j \mid \psi_j > 0} \text{NLL}(\hat{\sigma}, \hat{\xi}, \psi_j) \right] \quad (6)$$

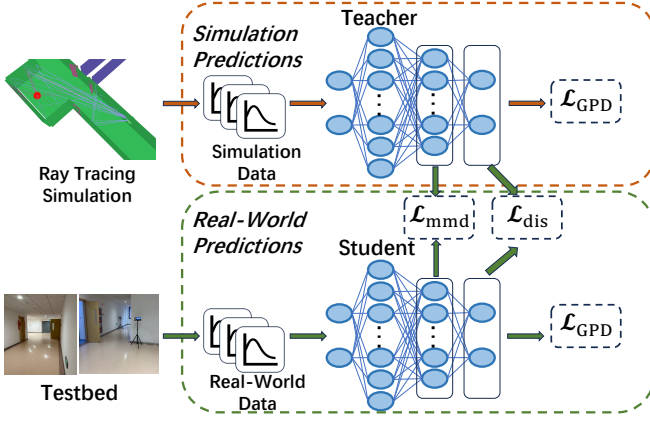


Fig. 2: Domain adaptation network architecture.

where $\text{NLL}(\hat{\sigma}, \hat{\xi}, \psi_j) = \log \hat{\sigma} + (1 + 1/\hat{\xi}) \log(1 + \hat{\xi}\psi_j/\hat{\sigma})$, $\hat{y} = (\hat{\sigma}, \hat{\xi}) = f_{\theta_1}(x)$, and the excesses $\psi_j = f(\gamma_j) - \mu$ are derived from Γ and μ . This yields $\theta_1^* = \arg \min_{\theta_1} \mathcal{L}_{\text{teacher}}(\theta_1)$. In subsequent training, the teacher is *frozen* and produces pseudo-labels (the predicted parameters) $q(x^s) = f_{\theta_1^*}(x^s)$ for simulated inputs.

Student Initialization and Objective: Following parameter-initialized distillation, the student is initialized with the teacher weights, $\theta_2^{(0)} \leftarrow \theta_1^*$, and then optimized independently with the composite loss

$$\mathcal{L}(\theta_2) = \mathcal{L}_{\text{GPD}} + \alpha \mathcal{L}_{\text{dis}} + \beta \mathcal{L}_{\text{mmd}}, \quad (7)$$

where α and β balance the contributions of the three terms. Here, $\mathcal{L}_{\text{GPD}}$ enforces accuracy on real measurements, $\mathcal{L}_{\text{dis}}$ distills simulation knowledge by matching the teacher's pseudo-labels $q(x^s)$ on simulated samples, and $\mathcal{L}_{\text{mmd}}$ aligns latent feature distributions between simulation and real domains. The explicit forms of these losses are given in the next subsection.

C. Loss Function Design

1) GPD Negative Log-Likelihood Loss ($\mathcal{L}_{\text{GPD}}$): We adapt the Mean-Squared Error regression with a GPD NLL loss [15]. This approach does not regress on pre-estimated parameters but fits the *actual raw SINR samples* from the real domain D_r . Assume the real-domain dataset is $D_r = \{(x_i, \mu_i, \Gamma_i)\}_{i=1}^{N_r}$, where μ_i is the pre-calculated threshold (see Section II) and $\Gamma_i = \{\gamma_{i,j}\}$ is the set of raw SINR samples at location x_i . The student model $f_{\theta_2}(x_i)$ predicts the tail parameters $\hat{y}_i = (\hat{\sigma}_i, \hat{\xi}_i)$. The NLL loss is then computed over the excesses $\psi_{i,j}$ derived from Γ_i and μ_i :

$$\mathcal{L}_{\text{GPD}} = \mathbb{E}_{(x, \mu, \Gamma) \sim D_r} \left[\sum_{j | \psi_j > 0} \text{NLL}(\hat{\sigma}, \hat{\xi}, \psi_j) \right] \quad (8)$$

where $\hat{y} = (\hat{\sigma}, \hat{\xi}) = f_{\theta_2}(x)$, and the excesses $\psi_j = f(\gamma_j) - \mu$ are defined using the pre-calculated threshold μ (from the dataset) and the monotonic transform $f(\cdot)$ from Section II. This loss function ensures the student model learns the parameters that most accurately describe the tail of the real-world data.

2) Distillation Loss ($\mathcal{L}_{\text{dis}}$): The teacher network generates soft labels $q = f_{\theta_1}(x)$ using simulation-domain inputs. The student learns from these teacher-generated outputs to inherit cross-domain feature alignment and reduce overfitting on scarce real data:

$$\mathcal{L}_{\text{dis}} = \mathbb{E}_{x \sim D_s} [\|f_{\theta_2}(x) - q\|_2^2]. \quad (9)$$

This loss transfers knowledge embedded in the simulation domain to the student network, acting as a regularizer that stabilizes convergence and improves generalization. The weighting factor α controls the distillation intensity and is empirically tuned through validation experiments.

3) Domain-Consistent Loss ($\mathcal{L}_{\text{mmd}}$): Even after distillation, the hidden representations extracted from simulation and real data may not share the same feature distribution. To explicitly minimize the domain gap, we employ the Maximum Mean Discrepancy (MMD) [14] distance between the latent features of the two domains:

$$\mathcal{L}_{\text{mmd}} = \|\mathbb{E}_{x \sim D_s}[\phi(f_{\theta_1}(x))] - \mathbb{E}_{x \sim D_r}[\phi(f_{\theta_2}(x))]\|_2^2, \quad (10)$$

where $\phi(\cdot)$ denotes the mapping to a Reproducing Kernel Hilbert Space (RKHS). This term enforces feature-level consistency by aligning the high-order moments of the simulation and real distributions. The coefficient β modulates its influence and is selected to balance alignment stability and fitting accuracy.

D. Training and Inference Procedure

The teacher network is first trained with the simulation dataset D_s by minimizing the GPD NLL loss $\mathcal{L}_{\text{teacher}}(\theta_1)$ as defined in Eq. (6). After convergence, the student network is jointly trained with both simulation and real samples by minimizing the composite loss (7).

In implementation, both networks adopt an MLP [16] with 3 hidden layers (128 neurons each) activated by ReLU and a linear output layer with two neurons corresponding to (σ, ξ) .

Batch normalization is applied after each hidden layer to accelerate convergence. The teacher is trained with the Adam optimizer (learning rate 10^{-3}). The student is trained with the SGD optimizer (learning rate 10^{-3} , momentum 0.9) and a batch size of 256. The weighting factors are empirically set to $\alpha = 0.4$ and $\beta = 0.8$. Once trained, the student model predicts the GPD parameters at unseen spatial locations, which are then used in (5) (along with the pre-calculated $\mu(l)$) to compute the URLLC-required SINR threshold $\gamma_{\text{tar}}(l)$ across the entire coverage area.

IV. DATASET AND TESTBED

To validate our approach, we collect real-world data from a physical testbed and generate a corresponding simulation dataset. Our testbed is located in a geometrically complex indoor environment of approximately 40 m $\times$ 40 m, featuring connected corridors with diverse reflective surfaces (plaster-board, glass, metal). Key parameters for the radio setup and datasets are summarized in Table I.

For the real-world dataset (D_r), we place the UE at 150 distinct locations. At each point, the UE remains stationary,

Setting	Specification
BS Platform	srsRAN on PC + USRP B210 [1]
UE Platform	Quectel RM500 5G modem [2]
Carrier Frequency	3.5 GHz
Bandwidth	20 MHz
Subcarrier Spacing	30 kHz

TABLE I: Experimental testbed and radio configuration.

Estimation Error	$\rho(\sigma)$	$\rho(\xi)$	Target SINR (dB)
Simulation Only	-0.196	0.071	7.71
Gaussian Process	0.120	0.016	6.86
Domain Adaptation	0.905	0.210	3.02

TABLE II: Estimation errors and correlations at different target SINR levels.

and we record uplink SINR values for 20 minutes at a 1 Hz rate, capturing approximately 1200 samples per location to ensure sufficient data for tail analysis.

For the simulation dataset (D_s), we first reconstruct the physical environment in Blender, modeling walls and materials. This 3D model is imported into the Sionna-RT ray-tracing simulator. We place the transmitter at the same coordinates as the real BS and sample channel data at about 4200 locations on a 0.3 m grid. At each location, multiple channel realizations are computed to emulate temporal variability, creating a SINR sample list analogous to the real-world measurements.

V. RESULT

We evaluate our proposed DA model against two baselines: 1) **Simulation Only**: The teacher model f_{θ_1} trained solely on simulation data (using $\mathcal{L}_{\text{teacher}}$) and applied directly to real-world locations. 2) **Gaussian Process (GP)**: A non-parametric Gaussian Process regression model in [5] tested on the same simulation dataset D_s to predict GPD parameters.

A. Prediction Accuracy of Target SINR

We first quantitatively assess the model's ability to predict the correct GPD parameters and the resulting target SINR. Table II compares the performance of our DA student model against the baselines, evaluated on the real-world test set. We report the Pearson correlation coefficient (ρ) between the predicted and ground-truth GPD parameters (σ and ξ), and the Mean Absolute Error (MAE) of the final derived target SINR (in dB) required for 99.9% reliability. The results clearly demonstrate the failure of the simulation-only models. The Simulation Only model shows a massive 7.71 dB error and a negative correlation for the scale parameter σ , indicating it completely fails to capture the real-world channel statistics. The Gaussian Process performs slightly better but is still highly inaccurate (6.86 dB error) and shows near-zero correlation for the parameters.

In contrast, our DA model is highly effective. It reduces the SINR prediction error by 55% (from 6.86 dB to 3.02 dB) compared to the next-best baseline. More importantly, it

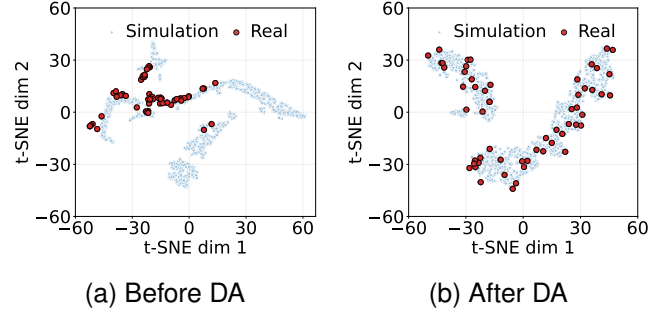


Fig. 3: t-SNE visualization of hidden features from the source-only and domain adaptation networks.

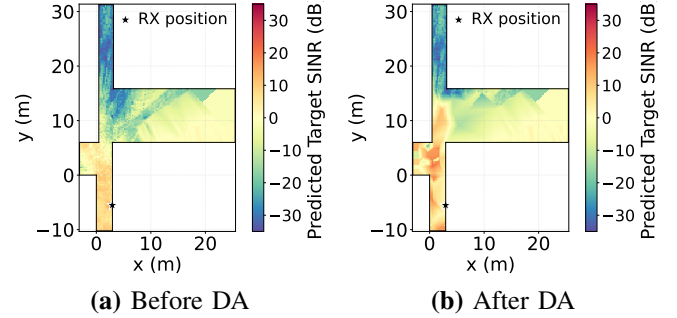


Fig. 4: Predicted SINR maps before and after domain adaptation.

achieves a very high correlation (0.905) for the scale parameter σ , proving that our model is learning the correct underlying statistical distribution of the tail, not just making arbitrary spatial predictions.

B. Domain Adaptation Visualization

To qualitatively verify that our model bridges the sim-to-real gap, we visualize the latent feature space using t-SNE [13] in Figure 3. The features are extracted from the penultimate layer of the networks.

Figure 3(a) shows the features from the Simulation Only model. The simulation samples and real-world samples form two distinct clusters. This visually confirms the significant domain gap between simulation and reality; a model trained on one cannot generalize to the other.

Figure 3(b) shows the features from our final DA model. The two distributions are now thoroughly intermixed, with the real-world samples spread throughout the simulation cluster. This demonstrates that our composite loss function, particularly the $\mathcal{L}_{\text{mmd}}$ term, successfully aligned the latent distributions and created a single, unified feature space for both domains.

C. Reliability Map Visualization

We now examine the practical output: the predicted spatial radio map. Figure 4 compares the target SINR map generated by the Simulation Only model (a) with our DA model (b). The maps show the predicted SINR (in dB) required at each location to achieve 99.9% reliability. The simulation domain

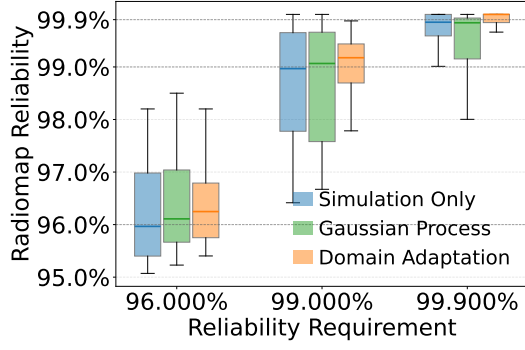


Fig. 5: Achieved vs. target reliability for different methods.

map in Figure 4 (a) is clearly incorrect. It exhibits a strong pessimistic bias, predicting that most of the environment, including the main corridor, requires a target SINR of 0 dB or less (green/blue). This is a direct consequence of the sim-to-real gap. Our DA map in Figure 4 (b) presents a much more realistic and spatially detailed picture. It correctly identifies high SINR zones (yellow/red, 10-20 dB) in the area with line-of-sight to the RX position and shows a gradual, physically plausible decay into the non-line-of-sight corridors. This map corrects the simulation bias and provides a trustworthy tool for 5G network reliability.

D. Radio Map Reliability

Finally, we evaluate whether the constructed reliability map truly delivers the promised reliability. For each location l in the real-domain test set, we first use the predicted GPD parameters to derive the target SINR $\gamma_{\text{tar}}(l)$ from the radio map. We then feed this $\gamma_{\text{tar}}(l)$ back into the real SINR trace collected at the same location and compute the empirical radiomap reliability as the fraction of samples whose SINR exceeds $\gamma_{\text{tar}}(l)$. We compare the required reliability (x-axis) against the actual reliability achieved by the map (y-axis) in Figure 5. The results are stark. At lower reliability requirements (e.g., 96% or 99%), all models perform adequately, with our DA model showing the tightest variance (i.e., it is the most dependable). However, at the 99.9% URLLC requirement, the baselines collapse. Both the Simulation Only and Gaussian Process models exhibit large variance, with reliability dropping as low as 98.0%. Their median performance is also below the 99.9% target. This means they are not ultra-reliable at all: a planner using these maps would fail to meet the URLLC service guarantee.

Our DA model achieves the most consistent performance among all baselines. Its predicted reliability values are tightly concentrated near the 99.9% requirement, with median performance meeting or exceeding the target and worst-case reliability remaining above 99.0%. These results suggest that our approach provides stable and reliable tail estimates suitable for stringent URLLC requirements.

VI. CONCLUSION

In this letter, we propose a novel domain-adaptive framework to construct high-fidelity, EVT-based reliability maps

for URLLC. Our approach successfully addresses two challenges: sim-to-real gap and the scarcity of extreme events. By employing a domain adaptive teacher-student architecture trained with a GPD NLL loss, our model learns to accurately predict the SINR tail distribution from raw channel samples. We validated our method using a real-world indoor testbed and a corresponding ray-tracing simulation. The results demonstrate that our framework effectively bridges the domain gap—reducing the 99.9% target SINR prediction MAE by over 55% compared to baselines. More critically, our model was the only one to remain robust and accurate at the 99.9% reliability requirement, making it a viable tool for practical URLLC network deployment.

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